



Crash Course Computer Science

1. Early Computing

- 算盘 → 步进计算器 → 差分机 → 分析机 → 打孔卡片制表机

2. Electronic Computing

- 继电器(relay) → 真空管(vacuum tube) → 晶体管(transistor)

3. Boolean Logic & Logic Gates

- Not, And, Or, XOR

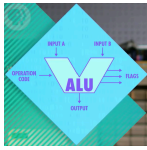
4. Binary

- 1 byte = 8 bits
- Use the first bit for the sign (1 for negative; 0 for positive)
- ASCII → Unicode

5. ALU (Algorithm & Logic Unit)

- arithmetic unit
 - half adder
 - full adder
 - 8-bit ripple carry adder
 - Carry-Look-Ahead adder

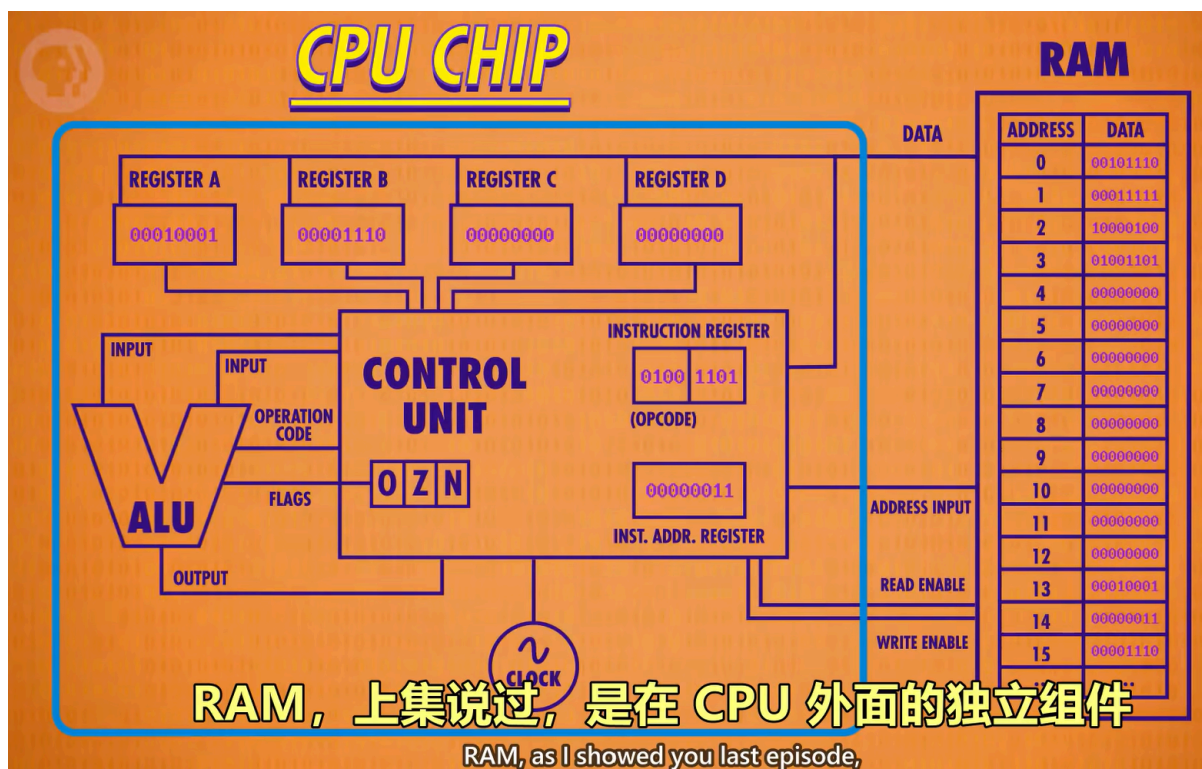
- logic unit
- illustration:



6. Registers and RAM

- latch
 - AND-OR latch
 - Gated latch
- Register: a group of latches
 - can hold a single number
 - The number of bits in a register is called its width
 - matrix for more bits with memory address
 - multiplexer: To convert from an address into something that selects the right row or column.
- Random Access Memory 随意存取存储器
 - Can access any memory location, at any time, and in a random order.

7. CPU (The Central Processing Unit)

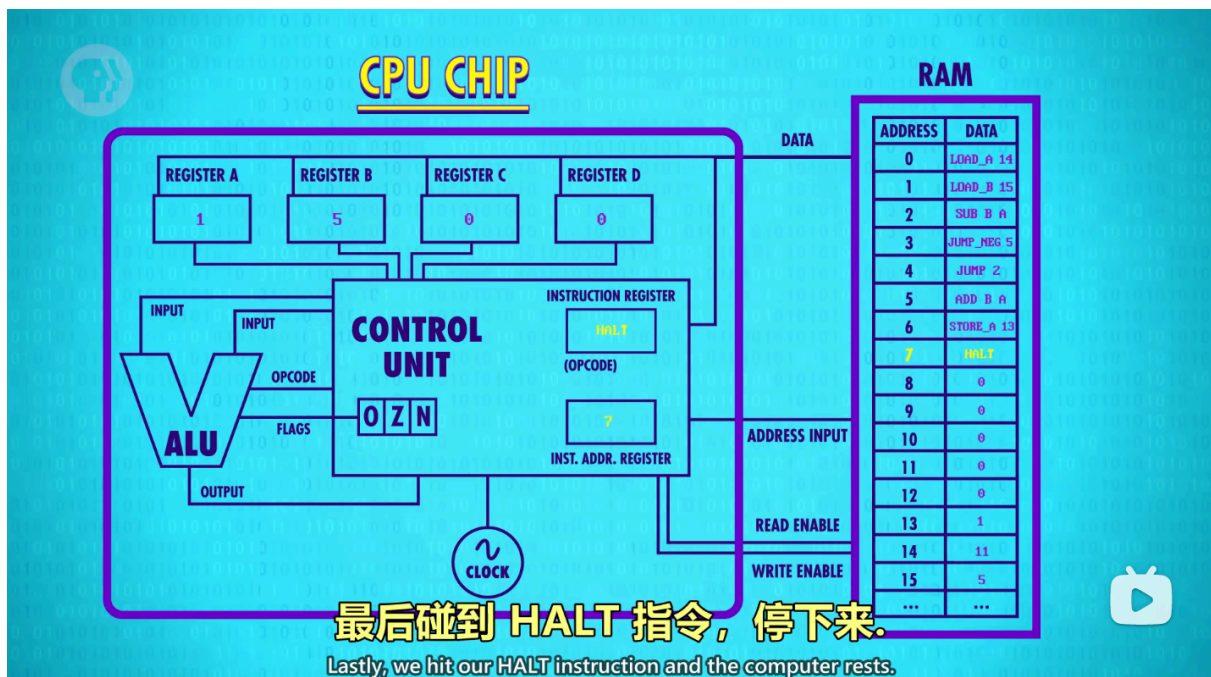


- 概念：
 - 寄存器+控制单元+ALU+时钟
 - 与RAM配合执行程序
 - CPU和RAM之间用“地址线”、“数据线”和“允许读/写线”通信
- 控制单元重要组件：
 - 指令表：给指令分配ID
 - 指令寄存器：存储当前指令
 - 指令地址寄存器：存储当前指令的内存地址
- 执行过程：
 1. 取指令 Fetch
 2. 解码 Decode
 3. 执行 Execute
- 时钟 Clock
 - 超频：加快CPU速度，可能过热或乱码
 - 降频：省电

- 动态调整频率

8. Instructions and Programs

- 指令集：记录指令名称、用法、操作码以及所需RAM地址位数
- 常用指令：
 - LOAD：数据从RAM到寄存器
 - STORE：数据从寄存器到RAM
 - ADD
 - SUB
 - JUMP
 - 有条件JUMP如JUMP-negative
 - HALT：停止
- 扩展指令长度：
 - 更多位
 - 可变指令长度
- 案例：



9. Advanced CPU Designs

- 减少晶体管切换时间
- 设计复杂电路: complexity vs speed
- 在CPU内设置缓存
 - 缓存命中/非命中
 - 脏位: 缓存与RAM不一致的数据
 - 缓存同步: 缓存已满时处理脏位
- 指令流水线
 - 并行执行
 - 数据依赖性? 乱序执行
 - 条件跳转? 推测执行、分支预测
- CPU多核、计算机多CPU

10. Early Programming

- Stored-program computers: store programs and data
 - e.g. Von Neumann Architecture
- plug boards → punch cards → panel programming

11. The First Programming Languages

- 二进制 → 助记符 (汇编器) → A-0 (编译器) → FORTRAN
- machine code
- assembly code using assembler (汇编器)
- high-level programming code using compiler (编译器)

12. Programming Basics - Statements and Functions

- conditional statement: if
- loop: while; for
- function

13. Intro to Algorithms

- Big O Notation: complexity

- selection sort: $O(n^2)$
- merge sort: $O(n * \log n)$
- Dijkstra 算法

14. Data Structure

- Array 数组
- Struct 结构体
- Linked List 链表

```
struct Node
    variable i
    pointer next //指针
```

- Queue 队列
 - FIFO: first in first out
 - enqueue 入队 dequeue 出队
- Stack 栈
 - LIFO: last in first out
 - push 入栈 pop 出栈
- Tree 树: 单向

```
struct Node
    variable i
    pointer nextleft
    pointer nextright
```

- root 根节点 leaf 叶节点
- Graph 图: 可循环

15. Alan Turing

- Turing Machine
 - State variable + rules + read/write head

16. Software Engineering

- Object Oriented Programming
 - object: related functions/code
 - e.g. C++ C# Java Python
- API: Application Programming Interface
- IDE: integrated development environment 集成开发环境
- Source Control 源代码管理 e.g. git
 - 代码存在中心服务器
- QA: Quality Assurance testing 质量保证测试

17. Integrated Circuits & Moore's Law

- ICs: a group of components
- PCB(印刷电路板): printed circuit board
- chip technology:
 - 光刻、氧化、参杂...
- Moore's Law: the number of transistors on a microchip doubles approximately every two years

18. Operating Systems (OS)

- 有操作硬件的特殊权限，可以运行和管理其它程序
- 一般是开机第一个启动的程序，其他所有程序都由操作系统启动
- 软件和硬件之间的媒介
- virtual memory → dynamic memory allocation
- Unix操作系统
 - kernel内核：核心功能，如内存管理、多任务输入/输出
 - 其他工具

19. Memory & Storage

- memory: non-permanent, volatile易失性 e.g. RAM
- storage存储器: non-volatile

- delay line memory: sequential, 1 bit at one time
- magnetic core memory
- magnetic tape → hard disc drive
- compact disc (CD): reflecting light
- solid state drive (SDD): no need to move, fast

20. Files & File Systems

- file format
 - .txt .wav .bmp
- file system (using directory)
 - flat file system
 - hierarchical file system

21. Compression

- lossless compression
 - run-length encoding - removing redundancies
 - dictionary coders - Huffman Tree
 - e.g. GIF PNG PDF ZIP
- lossy compression
 - perceptual coding
 - e.g. lossy audio compression: MP3

22. Keyboards & Command Line Interfaces

- human-computer interaction: I/O
- keyboard (QWERTY)
- command line interfaces 命令行界面

23. Screens & 2D Graphics

- Cathode Ray Tubes (CRTs) 阴极射线管
 - vector scanning

- raster scanning 光栅扫描
- Liquid Crystal Displays (LCDS) 液晶显示器
- Character Generator: a graphic card 显卡
 - Read Only Memory (ROM) 只读存储器: store graphics for each character, called a dot matrix pattern
 - Screen Buffer
- Sketchpad: an interactive graphical interface
 - offer Computer-Aided Design (CAD)
 - earliest complete graphical application

24. the Cold World and Consumerism

- history

25. the Personal Computer Revolution

- interpreter 解释器 vs compiler 编译器
 - interpreter: interpret while the program runs
 - compiler: compile before the program runs
- IBM Compatible: users can use the huge system of hardware and software

26. Graphical User Interface (GUI)

- point & click interface
- WIMP Interface
 - windows, icons, menus and a pointer
- Event-driven Programming

27. 3D Graphics

- 3D projection
 - wireframe rendering 线框渲染
 - orthographical projection 正交投影
 - perspective projection 透射投影
 - scanline rendering 扫描线渲染 : to fill graphics

- Anti Aliasing 抗锯齿
- Painter's Algorithm: sort distance to determine occlusion
- Z-Buffering
- Back-Face Culling 背面剔除
- shading 阴影
 - flat shading
 - Gouraud Shading
 - Phong Shading
- Texture
- GPU: Graphic Processing Unit 图形处理单元

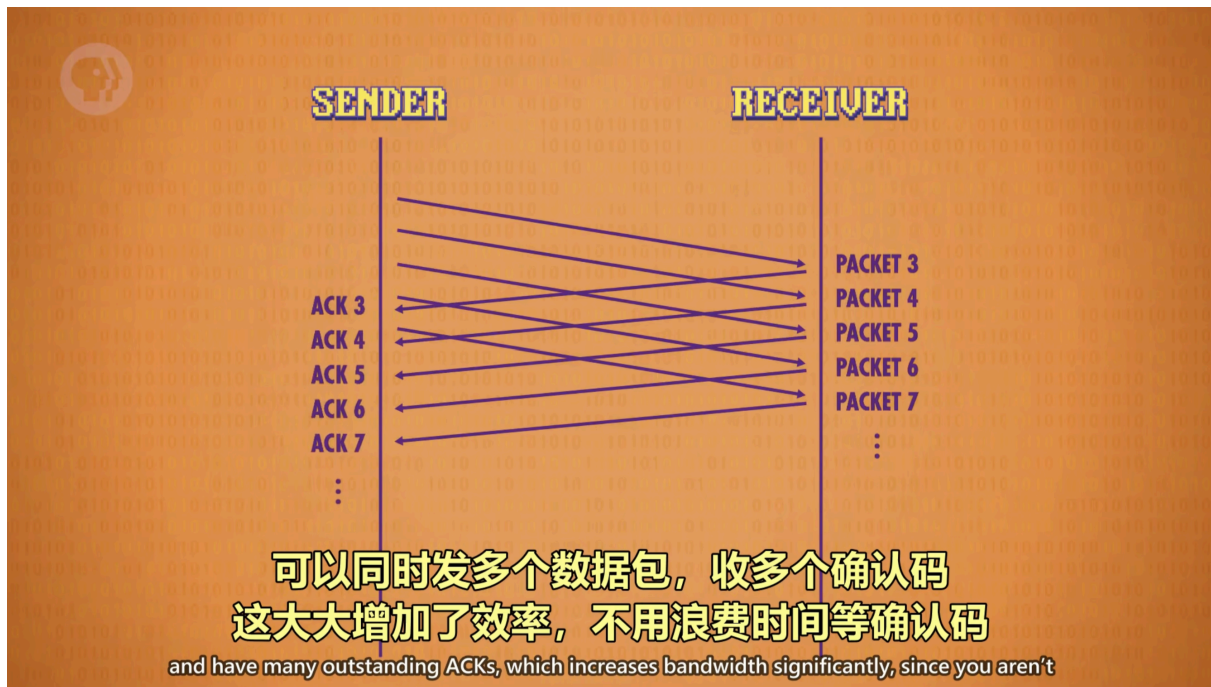
28. Computer Networks

- LAN: Local Area Network 局域网
 - Ethernet 以太网
 - MAC Address: Media Access Control Address, unique to every computer
- CSMA: Carrier Sense Multiple Access 载波侦听多路访问
 - Ethernet - copper wire
 - WiFi - air carrying radio waves
 - Bandwidth: the rate at which a carrier can transmit data
- Switch 交换机: connecting Collision Domains
- routing 路由
 - hop count 跳数
 - packets 数据包: to chop up big transmissions into many small pieces, called packets
- IP: Internet Protocol
 - defining the format of packet transmission
 - IP address

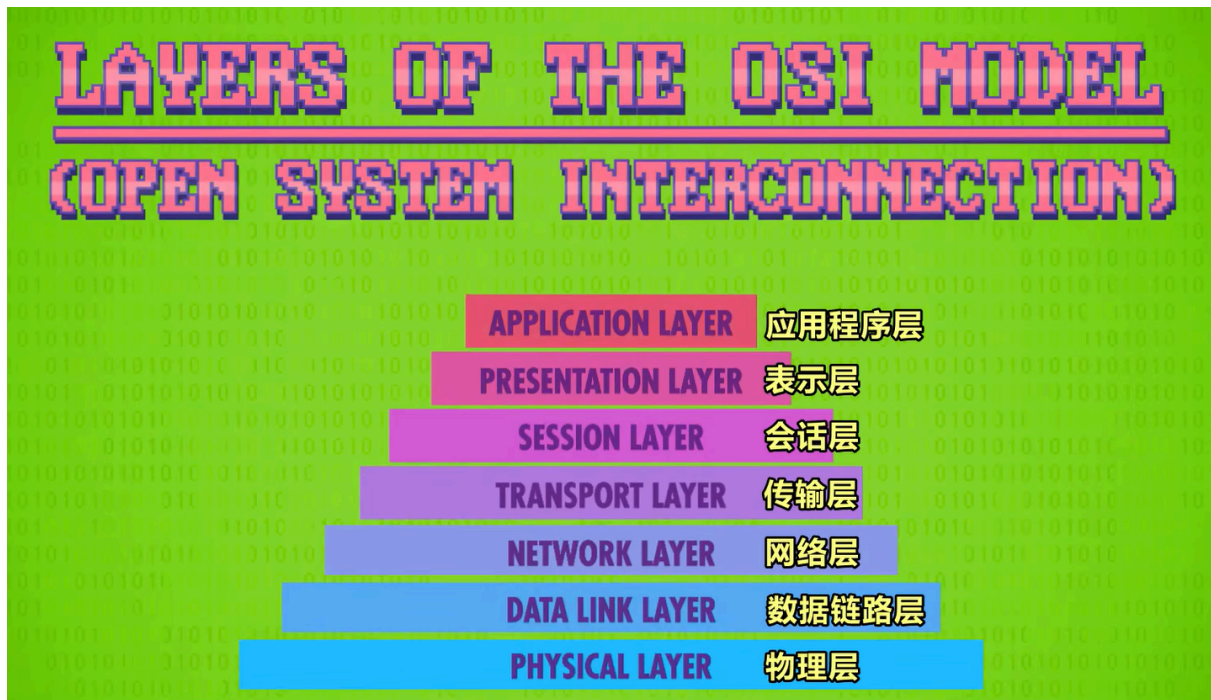
29. the Internet

- WAN: Wide Area Network 广域网

- belongs to ISP (Internet Service Provider)
- UDP: User Datagram Protocol 用户数据报协议
 - IP gets the packet to the right computer
 - UDP gets the packet to the right program running on that computer (port 端口)
- TCP: Transmission Control Protocol 传输控制协议 - TCP/IP



- DNS: Domain Name System 域名系统
- layers of the OSI model:



30. the World Wide Web

- Hyperlink 超链接
- URL: Uniform Resource Locator 统一资源定位器
- HTTP: Hypertext Transport Protocol 超文本传输协议
- HTML: Hypertext Markup Language 超文本标记语言
- Net Neutrality 网络中立性

31. Cybersecurity

- Secrecy 保密性, Integrity 完整性, Availability 可用性
- Authentication 身份验证
 - what you know: e.g. username, password
 - what you have: e.g. physical key
 - what you are: e.g. fingerprint
- Access Control 访问控制
 - Bell LaPadula model: ban reading up or writing down

32. Hackers & Cyber Attacks

- common methods

- Phishing 钓鱼
- Pretexting 假托
- Trojan Horses 木马
- Nand Mirroring Nand 镜像
- Exploit 漏洞利用
 - Buffer Overflow 缓冲区溢出
 - Code Injection 代码注入
- Worms 计算机蠕虫 - Botnet 僵尸网络
- DDoS 拒绝服务攻击

33. Cryptography

- cryptography 密码学
 - cipher 加密算法
 - ciphertext 密文
 - encryption 加密
 - decryption 解密
- defense in depth
- Substitution Cipher 替换加密
- Permutation Cipher 移位加密
- Data Encryption Standard 数据加密标准
- Advanced Encryption Standard 高级加密标准
- Key Exchange 密钥交换
- Symmetric Encryption 对称加密(密钥一样) & Asymmetric Encryption

34. Machine Learning & Artificial Intelligence

- Neural Network
- Reinforced Learning - narrow AI

35. Computer Vision

- Convolution 卷积 with Kernel 核

- Convolution Neural Network

36. Natural Language Processing

- Phrase structure rules 短语结构规则
- Speech Recognition 语音识别
- Phoneme 音素
 - recognition - break down into phonemes
 - synthesis - combine phonemes

37. Robots

- CNC machines: Computer Numerical Control 计算机数控
- controller
 - negative loop
 - PID(Proportional–integral–derivative) controller
- Azimov's Laws

38. Psychology of Computing

- usability 易用性
- affordance 直观功能

39. Educational Technology

- MOOC: Massive Open Online Course 大型开放式在线课程

40. the Singularity, Skynet and the Future of Computing

- ubiquitous computing: 普适计算
- singularity 奇点